Assignment no. 2

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Problem Statement : SMS Spam Collection

Data set:

label	message	length	word_count	label_encoded
ham	Go until jurong point crazy Available only in bugis n great world la e buffet Cine there got amore wat	102	20	0
ham	Ok lar Joking wif u oni	23	6	0
spam	Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005 Text FA to 87121 to receive entry	100	23	1
ham	U dun say so early hor U c already then say	43	11	0
ham	Nah I dont think he goes to usher no more	41	10	0
spam	FreeMsg Hey there darling it's been 3 week's now and no word back I'd like some fun you up for it still Tb ok XxX std chgs to send 150p/rec	139	30	1
ham	Even my brother is not like to speak with me They are not going to speak to me long But now gonna break that silence	116	25	0
ham	As per your request Melle Melle Oru Minnaminunginte Nurungu	138	25	0

	Vettam has been set as your call waiting tone for 15 days Try and change daily			
spam	WINNER You have been selected as a prize winner Text YES to 85023 now Standard rate applies	91	17	1
ham	Had your mobile 10 mths Update to latest Camera/Video phones for FREE See your 1 options	88	16	0
spam	SIX chances to win CASH From 100 to 20000 pounds txt CSH11 and send to 87575 Cost 150ppm	88	18	1
spam	URGENT Your Mobile No 07808726822 was awarded a 1500 Bonus Caller Prize to claim call 09050000327 NTT/THE/WKLY/BDG/OPT	118	17	1
ham	I've been searching for the right words to thank you for this breather I promise I wont take your help for granted and will fulfil my promise	141	27	0
ham	I HAVE A DATE ON SUNDAY WITH WILL	33	8	0
spam	XXXMobileMoviClub To use your credit click the WAP link in the next txt message or click here	93	17	1
ham	Oh k I was actually sleeping and you woke me up	47	11	0
ham	Fine if that's the way u feel That's the way its gota b	55	13	0
spam	England v Macedonia dont be late for the important Euro 2004 match	125	24	1

	updates send GET to 89004 now 5p 4info Reply STOP to 87239			
ham	Is that seriously how you spell his name	40	8	0
ham	I'm going to try for 2 months ha	32	8	0

Operations on data set using NumPy and Pandas :

1) Conversion of length C word_count coloum into array

Input:

```
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
length_arr = df['length'].to_numpy()
word_arr = df['word_count'].to_numpy()
print(length_arr)
print(word_arr)
```

Output:

```
[102  23 100  43  41 139 116 138  91  88  88 118 141  33  93  47  55 125
  40  32]
[20  6 23 11 10 30 25 25 17 16 18 17 27  8 17 11 13 24
 8  8]
```

2) Print mean value of length and word_count

Input:

```
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
length_arr = df['length'].to_numpy()
word_arr = df['word_count'].to_numpy()
print("Mean length:", np.mean(length_arr))
print("Mean word_arr:", np.mean(word_arr))
```

Output:

```
Mean length: 82.65
Mean word_arr: 16.7
```

3) Print max C min value of the length and word_count

Input:

```python
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
length_arr = df['length'].to_numpy()
word_arr = df['word_count'].to_numpy()
print("Max length:", np.max(length_arr))
print("Max word_count:", np.max(word_arr))
print("Min length:", np.min(length_arr))
print("Min word_count:", np.min(word_arr))
```

Output:

```
Max length: 141
Max word_count: 30
Min length: 23
Min word_count: 6
```

4) Find the standard deviation for the length and word_count:

Input:

```python
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
length_arr = df['length'].to_numpy()
word_arr = df['word_count'].to_numpy()
print("Standard deviation:", np.std(length_arr))
print("Standard deviation:", np.std(word_arr))
```

Output:

```
Standard deviation: 39.02470371444221
Standard deviation: 7.043436661176133
```

5) Print the total word in word_count

Input:

```python
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
length_arr = df['length'].to_numpy()
word_arr = df['word_count'].to_numpy()
print("Total words:", np.sum(word_arr))
```

Output:

```
Total words: 334
```

6) Identify all SMS messages whose length is greater than 100 characters using NumPy conditional operations.

Input:

```
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
length_arr = df['length'].to_numpy()
long_msgs = length_arr[length_arr > 100]
print("Long messages:", long_msgs)
```

Output:

```
Long messages: [102 139 116 138 118 141 125]
```

7) Sorting of array using numpy

Input:

```
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
length_arr = df['length'].to_numpy()
sorted_length = np.sort(length_arr)
print(sorted_length)
```

Output:

```
[ 23  32  33  40  41  43  47  55  88  88  91  93 100 1
02 116 118 125 138
 139 141]
```

8) Count the the no of time 1 is in label_encoded using numpy

Input:

```
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
label_arr = df['label_encoded'].to_numpy()
print(label_arr)
print("Count of 1:", np.count_nonzero(label_arr == 1))
```

Output:

```
[0 0 1 0 0 1 0 0 1 0 1 1 0 0 1 0 0 1 0 0]
Count of 1: 7
```

9) count the no. of spam message

input:

```
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
label_arr2 = df['label'].to_numpy()
print(label_arr2)
print("Count of spam messages:", np.sum(label_arr2 == 'spam'))
```

Output:

```
['ham' 'ham' 'spam' 'ham' 'ham' 'spam' 'ham' 'ham' 'sp
am' 'ham' 'spam'
 'spam' 'ham' 'ham' 'spam' 'ham' 'ham' 'spam' 'ham' 'h
am']
Count of spam messages: 7
```

10) Compare spam and ham messages in the dataset using NumPy operations.

Input:

```
import pandas as pd
import numpy as np
df = pd.read_excel('sms_spam_collection.xlsx')
spam_lengths = df[df['label'] == 'spam']['length'].to_numpy()
ham_lengths = df[df['label'] == 'ham']['length'].to_numpy()

print("Spam avg length:", np.mean(spam_lengths))
print("Ham avg length:", np.mean(ham_lengths))
```

Output:

```
Spam avg length: 107.71428571428571
Ham avg length: 69.15384615384616
```

11) Print the first and last 5 rows of the data set

Input:

```
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print("First 5 rows:", df.head(5))
print("last 5 rows:", df.tail(5))
```

Output:

```
First 5 rows:   label  ... label_encoded
0   ham  ...             0
1   ham  ...             0
2  spam  ...             1
3   ham  ...             0
4   ham  ...             0

[5 rows x 5 columns]
last 5 rows:    label  ... label_encoded
15   ham  ...             0
16   ham  ...             0
17  spam  ...             1
18   ham  ...             0
19   ham  ...             0

[5 rows x 5 columns]
```

12) Display the info of the dataset :

Input:

```
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print("information:\n",df.info())
```

Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   label          20 non-null     str
 1   message        20 non-null     str
 2   length         20 non-null     int64
 3   word_count     20 non-null     int64
 4   label_encoded  20 non-null     int64
dtypes: int64(3), str(2)
memory usage: 932.0 bytes
information:
```

13) Display the describe of the dataset:

Input:

```
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print("information:\n",df.describe())
```

Output:

```
information:
           length  word_count  label_encoded
count   20.000000   20.000000       20.00000
mean    82.650000   16.700000        0.35000
std     40.038501    7.226414        0.48936
min     23.000000    6.000000        0.00000
25%     42.500000   10.750000        0.00000
50%     89.500000   17.000000        0.00000
75%    116.500000   23.250000        1.00000
max    141.000000   30.000000        1.00000
```

14) Filter the data as word_count > 20

Input:

```
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print("word_count > 20\n" , df[df["word_count"] > 20])
```

Output:

```
word_count > 20
    label  ... label_encoded
2   spam  ...             1
5   spam  ...             1
6    ham  ...             0
7    ham  ...             0
12   ham  ...             0
17  spam  ...             1

[6 rows x 5 columns]
```

15) Add new column of status

Input:

```
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
df["status"] = ["seen" , "unseen" , "unseen" , "unseen" ,"unseen",
                "unseen","unseen","unseen","unseen","seen","seen",
                "unseen" , "unseen" , "unseen" ,"unseen", "unseen"
               ,"unseen","unseen","unseen","seen"]
print(df)
```

Output:

```
   label  ...  status
0    ham  ...    seen
1    ham  ...  unseen
2   spam  ...  unseen
3    ham  ...  unseen
4    ham  ...  unseen
5   spam  ...  unseen
6    ham  ...  unseen
7    ham  ...  unseen
8   spam  ...  unseen
9    ham  ...    seen
10  spam  ...    seen
11  spam  ...  unseen
12   ham  ...  unseen
13   ham  ...  unseen
14  spam  ...  unseen
15   ham  ...  unseen
16   ham  ...  unseen
17  spam  ...  unseen
18   ham  ...  unseen
19   ham  ...    seen

[20 rows x 6 columns]
```

16) Rename the column label to type

Input:

```
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print(df.rename(columns={'label': 'type'}))
```

Output:

```
    type  ...  label_encoded
0    ham  ...              0
1    ham  ...              0
2   spam  ...              1
3    ham  ...              0
4    ham  ...              0
5   spam  ...              1
6    ham  ...              0
7    ham  ...              0
8   spam  ...              1
9    ham  ...              0
10  spam  ...              1
11  spam  ...              1
12   ham  ...              0
13   ham  ...              0
14  spam  ...              1
15   ham  ...              0
16   ham  ...              0
17  spam  ...              1
18   ham  ...              0
19   ham  ...              0

[20 rows x 5 columns]
```

17) Calculate the average message length for each category (spam and ham) using Pandas groupby operation.

Input:

```
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print(df.groupby('label')['length'].mean())
```

Output:

```
label
ham      69.153846
spam    107.714286
Name: length, dtype: float64
```

18) Using Pandas, determine the frequency of each message category (spam and ham) in the dataset.

Input:

```python
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print(df['label'].value_counts())
```

Output:

```
label
ham      13
spam      7
Name: count, dtype: int64
```

19) Using Pandas, identify the distinct values in the 'label' column of the dataset.

Input:

```python
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print(df['label'].unique())
```

Output:

```
<StringArray>
['ham', 'spam']
Length: 2, dtype: str
```

20) Remove the 'label_encoded' column from the dataset using Pandas

Input:

```python
import pandas as pd
df = pd.read_excel('sms_spam_collection.xlsx')
print(df)
df.drop("label_encoded", axis=1, inplace=True)
print("after deleting label_encoded coloumn\n", df)
```

Output:

```
after deleting label_encoded coloumn
   label                                            message  length  word_count
0    ham  Go until jurong point crazy Available only in ...     102          20
1    ham                            Ok lar Joking wif u oni      23           6
2   spam  Free entry in 2 a wkly comp to win FA Cup fina...     100          23
3    ham        U dun say so early hor U c already then say      43          11
4    ham          Nah I dont think he goes to usher no more      41          10
5   spam  FreeMsg Hey there darling it's been 3 week's n...     139          30
6    ham  Even my brother is not like to speak with me T...     116          25
7    ham  As per your request Melle Melle Oru Minnaminun...     138          25
8   spam  WINNER You have been selected as a prize winne...      91          17
9    ham  Had your mobile 10 mths Update to latest Camer...      88          16
10  spam  SIX chances to win CASH From 100 to 20000 poun...      88          18
11  spam  URGENT Your Mobile No 07808726822 was awarded ...     118          17
12   ham  I've been searching for the right words to tha...     141          27
13   ham                    I HAVE A DATE ON SUNDAY WITH WILL      33           8
14  spam  XXXMobileMoviClub To use your credit click the...      93          17
15   ham    Oh k I was actually sleeping and you woke me up      47          11
16   ham  Fine if that's the way u feel That's the way i...      55          13
17  spam  England v Macedonia dont be late for the impor...     125          24
18   ham             Is that seriously how you spell his name      40           8
19   ham                      I'm going to try for 2 months ha      32           8
```